EMILIO GUADARRAMA GUTIÉRREZ

MECHATRONICS ENGINEERING STUDENT – ELECTRONICS & EMBEDDED SYSTEMS

ABOUT ME

I am a Mechatronics Engineering student at Tecnológico de Monterrey with a strong focus on electronics, embedded systems, and PCB design. My experience in robotics, rocketry, and leadership roles has helped me develop analytical thinking, teamwork, and innovation skills. I aim to contribute to multidisciplinary projects that merge technology, automation, and design.

EDUCATION

Tecnológico de Monterrey (2025- 2029)

B.S. in Mechatronics Engineering

LANGUAGES

English: Full Professional Proficiency
Spanish: Native Speaker
German: Elementary proficiency

TECHNICAL SKILLS

- **PCB Design**
- **Embedded Systems**
- **Altium Designer**
- **Python**
- **React App**
- **ESP32**

C, C++, Python, Arduino, Git and IoT systems

CONTACT

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EXPERIENCE

Ignitia Rocket Lab – Electronics Team Member

Aug 2025 – Present | Zapopan, Jalisco

Contributing to the electronics and avionics division of a competitive rocketry team representing Tecnológico de Monterrey.

- Focused on PCB design, embedded systems, and control electronics.
- Supporting the development of ignition and data acquisition systems.
- Collaborate with multidisciplinary teams across propulsion, structures, and systems integration.

FIRST Robotics Competition – Team Daedalus

Jan 2023 – May 2025 | Zapopan, Jalisco

- **Mechanical Lead & Mentor (2025 Season)** – Guided the team's final high-school season, mentoring new members and participating in Monterrey and León Regionals.
- **Mechanical Lead (2024 Season)** – Led mechanical area and drive team; qualified for the **FIRST World Championship in Houston** by winning the Regional Hermosillo.
- **Robotics Technician (2023 Season)** – Contributed to scouting, drive team, and mechanical systems operations.

TECMUN GDL – Under-Secretary-General

Aug 2024 – Jun 2025 | Zapopan, Jalisco

Served as Under-Secretary-General for the 2024 edition of PrepaTec's Model United Nations.

- Led a 10-member Secretariat to organize a conference with 500+ delegates and 100 staff members.
- Coordinated logistics, supervised debate protocols, and managed communication with faculty.
- Strengthened skills in leadership, organization, and public speaking.

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MECHATRONICS ENGINEERING

PROJECTS

Flight Computer – Ignitia Rocket Lab

Team Project: Ignitia Rocket Lab (Electronics Division)
Description:
Contributed to the reverse engineering and redesign of a flight computer based on an ESP32-Devkit V1. Responsible for PCB layout and system integration in Altium Designer, including e-match control and telemetry communication.
Technologies: Altium Designer, ESP32, AMS1117 Regulators and PCB Manufacturing

Smart irrigation system – Water Growth Solutions (WGS)

Team Project: Sustainable Technology & Renewable Energy Initiative
Description
Collaborated in a multidisciplinary team to create WaterGrowth Solutions (WGS), a sustainable irrigation and vertical garden system for rural communities.
I was responsible for the Arduino code, electronic control system, and web interface (React), enabling real-time monitoring and automation.
Other teammates developed the mobile app, sustainability framework (UN SDGs), and documentation.
Aimed to improve irrigation efficiency and food security in rural communities through sustainable automation.
Technologies: Arduino, React, IoT, Bluetooth/Wi-Fi Communication, Web Interface and Solar Energy

CERTIFICATIONS

TOEFL iBT (Score: 98, C1 CEFR) – ETS

Issued April 2025
Skills: Academic English, CEFR C1 Level

Advanced Robotics Course – Asociación de Robótica Aplicada y Ciencias de la Tecnología (UNAM)

Issued July 2024
Duration: 45 hours | Final Grade: 10/10
Focus: Arduino IDE, Robotics, Automation

Honorific Mention - Bicultural High School

Tecnológico de Monterrey | May 2025
Awarded for academic excellence during the Bicultural High School Program, with a final **GPA of 96/100.**

VOLUNTEERING

TECHO Méxcio

Crew leader | Crew Co-leader | Volunteer
Oct 2024 – Present | Jalisco, Mexico

Participated in the construction of emergency housing for families in vulnerable communities.

- **Crew Leader** for the third construction in Comunidad de Jauja, Tonalá, Jalisco.
- **Crew Co-Leader** for the second construction in Comunidad Indígena Sergio Barrios, Tlajomulco, Jalisco.
- **Volunteer** during the first construction in Comunidad Indígena Sergio Barrios, Tlajomulco, Jalisco.

Strengthened teamwork, communication, and leadership skills while collaborating with local communities.